Research activity

The point of a line closest to a given point — 35 min.

In an orthonormal frame of space, consider $A(1,0,0)$, $M(0,2,3)$ and the line d through A with direction vector $\vec{u} = (1,1,0)$. We look for the point of d closest to M.

1. Express, in terms of a real parameter t, the coordinates of a point H of d.
2. Find the value of t for which $\overrightarrow{MH} \cdot \vec{u} = 0$. Deduce the coordinates of the resulting point H.
3. Calculate the distance MH.
4. Let P be any point of d. Prove that $MP^2 = MH^2 + HP^2$, then deduce that H is the unique point of d closest to M.